Sebastian Krantz

Senior Consultant at CPCS | Consultant at World Bank | Fellow at Kiel Institute

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Core Competencies

- Economic Research and Analysis in both Academic and Policy Contexts
- Strong Skills in Econometrics, Time Series, and Machine Learning, and Practical Experience with DSGE, CGE, Spatial Economic & Freight Models
- High-Performance Statistical Software Engineering and Full-Stack Web-Application and API Development
- Experience with Transport, Trade, and Infrastructure Analytics in Africa/EMDEs

Education

PhD in Quantitative Economics, Magna cum Laude, University of Kiel / Kiel Institute. 2021–2024
Dissertation: Africa's Economic Transformation: A Big Data Perspective (Keywords: Optimal Infrastructure Investments, Regional Integration, Macroeconomic Stabilization)
Supervisor: Prof. Tobias Heidland, *Second Reader:* Prof. Christoph Trebesch.

Master in International Economics (MIE), Geneva Graduate Institute (IHEID). 2017–2019
Focus: Macro, Development, Econometrics, Trade. **GPA**: 5.54/6 at 156/120 ECTS
Dissertation: Endogenous R&D and Technology Diffusion in a Multi-Sector RBC Economy (Keywords: DSGE, Endogenous Growth, Business Cycles, Sectoral Shocks)
Supervisor: Prof. Cédric Tille, *Second Reader*: Prof. Rahul Mukherjee.

BA. Liberal Arts and Science, Magna cum Laude, University College Roosevelt (UCR) - Liberal Arts & Sciences Honors College of Utrecht University. 2014–2017
Major: Economics and Social Science (Economics, Anthropology, IR, Law, History)
Minor 1: Mathematics, **Minor 2**: Statistics. **GPA**: 3.87/4 at 202.5/180 ECTS
Honors Thesis: The Interdependence of Economic Growth, Human Development, and Political Institutions: A General Equilibrium Framework

Employment

Senior Consultant (Analytics) at CPCS – Advisors to Infrastructure Leaders 2025-2026

CPCS is a global management consulting firm specializing in transport and energy sectors worldwide. I am part of their International Analytics Team. So far, I have lead multimodal freight modelling for Thailand Ministry of Transport's "Integrated Logistics and Intermodal Transport Development Plan", and assisted modelling work for the US Department of Transport. I also created the flownet package for transport modeling, and a Global Multimodal Freight Model (115k links) + AI Platform enabling rapid AI-assisted appraisal/evaluation of major projects. I additionally forecasted traffic along two road corridors in Cameroon, lead spatial analytics for agricultural site selection around a road corridor in Gabon, and developed and delivered a full-stack Public-Private Partnership (PPP) management platform for the PPP Coordination and Advisory Unit (UC-PPP) of the DRC. I am currently modelling Sierra Leone's power system to assess the impact of a $500M Power Sector Compact enhancing the transmission network.

PhD Researcher at the Kiel Institute for the World Economy 2021–2024

The Kiel Institute for the World Economy (IfW) is a leading research institute on global economic affairs. I was a junior researcher in the Research Center International Development and the Africa Initiative.

Research

Papers on macroeconomic stabilization (Africa's Great Moderation), EAC regional integration (both published), and economic returns to infrastructure/optimal infrastructure investments in Africa. My first infrastructure paper builds the granular Africa Infrastructure Database and uses causal ML to map local economic returns to incremental investments in 14 types of infrastructure at 9.7km resolution. The second characterizes economically optimal investments into Africa's road network in partial and general equilibrium – calibrating spatial models to a realistic topography of the continental road network.

Other Outputs

Developed the Africa Monitor (a macroeconomic information platform for Africa with R and Python APIs), an open macroeconomic database and nowcasting platform for South Africa, consulted GIZ/MoF Benin and the World Bank Africa Chief Economist's Office, and engaged in extensive software engineering.

Senior Economist (ODI Fellow) in the Macroeconomic Policy Department, Ministry of Finance, Planning and Economic Development, Uganda 2020-2021

Analytics & Research

Macroeconometric analysis of planned increase in excise duties on petrol and diesel. Econometric estimation of elasticity parameters for a CGE model to evaluate SDGs. Development of a GDP nowcast. COVID-19 tourism sector emergency response paper and strategy paper on cultural tourism development. Work with World Bank team to update adjusted macroeconomic indicators for Uganda and developed model to forecast them. Developed model to perform disaggregated annual revenue forecasting by tax head + short-term (monthly, quarterly) models. Started revision of department's main financial programming model. Research papers on the effects of fiscal spending (SVAR + multiplier analysis) and GVCs/RVCs in the EAC.

IT and Process Innovation

Automated monthly scraping of ~10,000 macroeconomic indicators from national sources + codes for >3000 national indicators. Developed UGATSDB: a mixed frequency time series database + open online data portal and R API package for easy data retrieval. The database facilitates analysis and powers a macroeconomic dashboard that displays the GDP nowcast and leading economic indicators. Automated the department's monthly economic performance report using computable documents linking to UGATSDB. Built a modern department website integrating these services and making accessible all publications/reports.

Teaching and Capacity Building

Structured R-based Data Science Training Programme consisting of 6 delivered 2-day intensive courses taught in monthly intervals. Followed actively by 2 department staff which now maintain the website, database, apps, and computable monthly report. 4+ external (UBOS) staff also attended.

Publications

Working Papers

Krantz, S. (2024). Mapping Africa's Infrastructure Potential with Geospatial Big Data and Causal ML. *Kiel Working Paper.* https://ssrn.com/abstract=4537867.

Krantz & Bougna (2026). Optimal Road Investments in the CEMAC Region. *World Bank (Pending Review).* https://ssrn.com/abstract=6397299.

Krantz, S. (2026). Energy and Digital Infrastructure Complementarities. *World Bank (Pending Review).*

Under Review

Krantz, S. (2026). Optimal Investments in Africa's Road Network. *Kiel Working Paper and World Bank Policy Research Working Paper.* https://ssrn.com/abstract=4893771.

Felbermayr, et al. (2025). On the Geographical Incidence of Trade Wars: Regional Outcomes across Europe and the Role of the Single Market. R&R at *Journal of Economic Geography*.

Published

Krantz, S. (2026). collapse: Advanced and Fast Statistical Computing and Data Transformation in R. *Journal of Statistical Software, 116*(1), 1-38. https://doi.org/10.18637/jss.v116.i01.

Krantz, S. (2025). Patterns of Global and Regional Integration in the East African Community. *Review of World Economics*, *161*, 151-230. https://doi.org/10.1007/s10290-024-00558-0.

Krantz, S. (2024). Africa's Great Moderation. *Journal of African Economies*, *33*(5), 515-537. Oxford University Press. https://doi.org/10.1093/jae/ejad021.

Krantz, S. (2017). The Interdependence of Economic Growth, Human Development and Political Institutions: A General Equilibrium Framework. *Student Undergraduate Research E-journal*, *3*.

Academic dissertations and unpublished research/policy papers see sebastiankrantz.com/research.

Software Engineering

I develop high-performance open-source statistical software, focusing on C/C++-based R packages:

High-Performance Statistical Computing Architecture

- collapse: Advanced and Fast Statistical Computing and Data Transformation in R – one of the largest R packages (>40k lines, 30k of which C/C++), benchmarked on linux and windows in **global top 5** fastest solutions for database-like operations (>>Spark). **2.5+ million** downloads.
- fastverse: An Extensible Suite of High-Performance and Low-Dependency Packages for Statistical Computing and Data Manipulation in R – easy discovery, loading, and management.
- Contributed to kit (Data Manipulation Functions in C) and data.table (Enhanced Data Frame).

Time Series and Panel Data Econometrics

- dfms: Dynamic Factor Models for R – efficient C/C++-based estimation supporting missing data.
- Contributed to fixest (Fast Fixed-Effects Estimation) and plm (Linear Models for Panel Data).

Spatial Analytics

- osmclass: Efficiently Classify Open Street Map Features – into functional/analytical categories
- Contributed to DGGRID/dggridR: Discrete Global Grids (for R) – Spatial Analysis Done Right

Transport Networks and Traffic Assignment

- OptimalTransportNetworks.jl: Optimal Transport Networks in Spatial Equilibrium – in Julia
- flownet: Efficient Network Processing, Route Enumeration, and Traffic Assignment in R

Global Value Chain Analysis

- GlobalValueChains.jl: Fast Global Value Chain Decomposition in Julia
- Significantly contributed to decompr: Global Value Chain Decomposition in R

Database API Packages

- ugatsdb, africamonitor (also in Python), and samadb (also in Python and Julia).

For more info see sebastiankrantz.com/software, github.com/SebKrantz, and sebkrantz.github.io/Rblog. I also develop full-stack apps mentioned elsewhere and listed at sebastiankrantz.com/data-products.

Consulting

Digital Vice Presidency, The World Bank (Washington DC) 2025-26

Tasks: Analyze large-scale digital infrastructure and hazard data to quantify connectivity gaps and damage risk.
Outputs so far: Created the 'Infrastructure Risk Analyzer' – a full-stack web-app for infrastructure risk analysis.

ITS GeoLab, The World Bank (Washington DC) 2025-26

Tasks: Create and automate a Global Infrastructure Database for all LMIC and explore geospatial AI applications.
Outputs so far: Developed LMIC database: 100M+ features in 60+ categories from 10+ sources for 123 LMICs.

Office of the Chief Economist, Europe and Central Asia, The World Bank (Remote) 2025

Outputs: Policy research paper on Investments to Overcome Infrastructure Misallocation and Improve Connectivity in the ECA Region + data work for paper on Aggregate Economic Gains from Road Improvements.

Trade Policy Research Group, Kiel Institute for the World Economy (Kiel) 2025

Outputs: Developed regionally extended MRIO tables for the Kiel Institute Trade Policy Evaluation (KITE) model – covering 288 European NUTS2 regions and 47 non-EU countries in 10 NACE sectors for years 2016-2020.

CEMAC Infrastructure Sector Assessment Program, The World Bank (Remote) 2024-25

Outputs: Economic modelling of optimal road network investments in the CEMAC region in partial and general equilibrium and evaluation of regional road projects – producing a policy research paper (slides) + spatial dataset.

Office of the Chief Economist, Africa Region, The World Bank (Remote) 2024-25

Outputs: 3 short background papers on "Africa's Regional and Global Integration" (GVC and RVC analysis, predictors of integration, and economic returns) for report "Integrating Africa Forward: From Threads to Hubs".

Ministry of Economy and Finance and GIZ Benin (Cotonou, Benin) 2023

Public Finance Reform Program to achieve the SDGs and Strengthen Revenue Mobilization.

Outputs: Workshop on government revenue forecasting, outlining process in Germany and presenting in detail forecasting models developed for Uganda. Attended by ~15 staff from the Ministry of Economy and Finance.

Presentations

2022 – PEGNet Conference (Kampala), German Development Economics (GDE) Conference (Stuttgart), IDOS Conference (Bonn), Federal Ministry of Finance (Berlin), useR'22 (Virtual).

2023 – BBL at Stellenbosch University (Stellenbosch), Bureau for Economic Research (x2) (Stellenbosch), SEM Conference (Milan), GDE Conference (Dresden), IfW WCA Conference (Berlin), MIASA Policy Conference (Accra), Oxford Summer School (Oxford).

2024 – Global Policy Lab (Berkeley), CSAE Conference (Oxford), The World Bank (GOST - Virtual), IfW Research Seminar (Kiel), Japan International Cooperation Agency (Virtual), BMWK (Berlin).

2025 – United Nations Economic Commission for Africa (Virtual), Annual World Bank Conference on Transport Economics (DC), IFPRI Research Seminar (DC), World Bank Development Data Group (DC), CSAE Conference (Oxford), Statistics Portugal (Virtual), Bank of Portugal (Porto).

Research Visits and Internships

Visiting Researcher at Stellenbosch University Economics Department 2023

Theme: Nowcasting the Economy and Measuring Inequality with Geospatial Data.

Outputs: Developed the South Africa Macroeconomic Database and the SA Nowcast. Both were presented to a broad audience at BER. Investigated the potential of geospatial big data to map inequality across space and time and compared estimates to the Spatial Tax Panel. Presented at the department's BBL seminar.

Intern in the Chief Economists Office, Oxford Policy Management, Oxford (UK) 2018

Project: Rapid Technological Change - Challenges and Opportunities.

Outputs: Contributed to country-specific case studies on Indonesia and Tanzania analyzing sectoral pattern of growth, industry structure, and the innovation system using national accounts and enterprise survey data.

Intern in the Research Department, Bangladesh Rural Advancement Committee (BRAC), Kampala (Uganda). Project: Secondary Education in Uganda - New Evidence. 2018

Outputs: Executed a study on secondary education: Proposal, survey design + enumerator training, data collection on 500 secondary schools across the country, data cleaning, data analysis, write-up of findings. Developed a web application for semi-automated exploratory analysis and geo-visualization of surveys.

Computer and Language Skills

ANALYTICS & PROGRAMMING

- R (Advanced + C-API)
- C/C++ (Intermediate)
- Python (Intermediate)
- SQL (Intermediate)
- STATA (Intermediate)
- Julia (Intermediate)
- Mathematica (Beginner)
- Matlab & Dynare (Beginner)
- GEMPACK/GAMS (Beginner)
- Linux Server + NGINX (Beginner)

OFFICE, DATA PRODUCTS, MEDIA

- MS Office & Latex (Intermediate)
- QGIS (Intermediate)
- Shiny Apps (Advanced)
- Plotly Dash Apps (Beginner)
- Simple (HTML & CSS) Websites
- R Packages (Advanced)
- Python Packages (Beginner)
- Julia Packages (Beginner)
- Photo: Lightroom & Photoshop
- Video: IMovie & Premiere Pro

LANGUAGES

- German (Native)
- English (Fluent)
- French (B1)
- Dutch (A2)
- Spanish (A1)

Other Skills Trainings

BREAD-IGC Virtual PhD Course on Urban Economics. *(9 Online Lectures)* 2024

Political Economy and Development Economics, University of Oxford. *(1 Week)* 2023

Financial Programming and Policies, Macroeconomic Forecasting, and Revenue Forecasting, IMF Institute for Capacity Building. *(MOOCs via edX)* 2020-2021

Practical CGE Modeling, CoPS - Victoria University. *(2 Week Online Course)* 2020

The Science and Art of DSGE Modeling, CIMS - University of Surrey. **Specialization**: Emerging Open Economies. *(5 Day Boot Camp)* 2018

Big Data in Macroeconomics and Finance, Kiel Institute for the World Economy (IfW). *(6 Day Intensive Summer School)* 2018

Self-studied several MOOCs/books on **Data Science, ML, and Algorithms (DSA)** 2018-2024

Academic References

- **Prof. Tobias Heidland** (Director of the RC International Development at IfW, Main Supervisor)
- **Prof. Rainer Thiele** (Director of the Kiel Institute Africa Initiative at IfW/Kiel Institute)
- **Prof. Christoph Trebesch** (Director of the RC International Finance and Macroeconomics at IfW)
- **Prof. Julian Hinz** (Director of the RC Trade Policy at IfW/Kiel Institute)
- **Dr. Niclas Moneke** (Associate Professor at University of Oxford [Infrastructure Economics])
- **Prof. Cédric Tille** (Professor of Economics at IHEID [Macroeconomics], Dissertation Supervisor)

Professional References

- **Dr. Albert Musisi** (Commissioner of the Macroeconomic Policy Department, Ministry of Finance, Planning and Economic Development of Uganda)
- **Dr. Theophile Bougna** (Senior Economist, Transport Global Practice, The World Bank)
- **Dr. Woubet Kassa** (Economist, Office of the Chief Economist, Africa Region, The World Bank)
- **Dr. Stevan Lee** (Chief Economist at Oxford Policy Management)